iJRASET
International Journal For Research in Applied Science and Engineering Technology



INTERNATIONAL JOURNAL FOR RESEARCH

IN APPLIED SCIENCE & ENGINEERING TECHNOLOGY

Volume: 14 **Issue:** IX **Month of publication:** September 2026

DOI:

www.ijraset.com

Call: 08813907089 | E-mail ID: ijraset@gmail.com

International Journal for Research in Applied Science & Engineering Technology (IJRASET)
ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 7.538
Volume 14 Issue IX Sep 2026- Available at www.ijraset.com

Healthorbit: An Integrated Digital Health Information Platform with a Framework for Context-Aware AI Assistance

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Abstract: ***Personal health information is increasingly generated and stored in digital and document-based forms, yet users may encounter fragmented access to medical reports, health history, medicines, symptoms, measurements, and other health information. This paper presents HealthOrbit, an integrated digital health information platform for managing and interacting with personal health information through a unified application environment. The platform provides authenticated access and medical-document processing functionality, including an OCR-based workflow for extracting information from supported documents. HealthOrbit incorporates the concept of Health Context, through which relevant and authorised user-provided health information can be made available to contextual AI assistance. The paper presents the system architecture, functional modules, document-processing workflow, data-flow model, implementation technologies, security and privacy considerations, and future extensibility of the platform. The system is designed to provide a unified environment for organising personal health information while establishing a foundation for context-aware interaction with health-related information.***
Index Terms: Digital Health, Personal Health Information, Medical Document Processing, OCR, Context-Aware AI.

I. INTRODUCTION

Digital technologies are increasingly embedded in the collection, exchange, and management of health information. The World Health Organisation identifies digital health and interoperability as important components of modern health-system transformation, while emphasising that digital solutions should be aligned with health needs and evidence [1], [2].

At the individual level, health information may exist in different forms, including laboratory reports, prescriptions, diagnostic documents, personal health records, medication information, measurements, and historical records. A systematic review of personal health records notes the importance of patient-centred access to health information and identifies integration, standards, functions, and architecture as recurring concerns in the field [3].

The practical difficulty is therefore not only the availability of digital health information, but also the ability to organise, retrieve, understand, and interact with information that belongs to the same individual. Medical documents add to the challenge because valuable information may remain embedded in scanned or photographed documents. Research has demonstrated that OCR pipelines can convert printed medical records into machine-readable information and can support the construction of structured personal health records [4].

Artificial intelligence introduces another interaction layer. Conversational agents have been investigated for a range of healthcare applications, but systematic reviews also emphasise the need for appropriate evaluation, safety considerations, and careful interpretation of effectiveness [5], [6]. Recent studies of conversational large language models similarly identify broad healthcare applications alongside concerns related to reliability, privacy, and misuse [7].

HealthOrbit addresses this combined information-management problem by providing a unified health-information environment in which users can access available health information and interact with supported health-document functionality through a common interface. The platform uses the concept of Health Context to represent information that may be made available to contextual AI components. This paper presents the developed system, its document/OCR workflow, system architecture, data-flow model, implementation technologies, security considerations, and future extensions.

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II. LITERATURE SURVEY

A. Personal Health Records and Digital Health

Personal health records (PHRs) have been studied as patient-centred systems that allow individuals to access and interact with their own health information. Roehrs et al. reviewed the PHR literature and identified data types, standards, architectures, functions, and integration challenges as important areas of research [3]. This work supports the broader motivation for a unified environment in which information from different sources can be organised around an individual.

The World Health Organisation's digital-health strategy further emphasises digital technologies, interoperability, data sharing, and evidence-based implementation as important components of digital transformation in health [1], [2]. These principles are relevant to HealthOrbit's objective of integrating additional sources of health information while maintaining user-centred access.

B. Medical Document Processing and OCR

Medical information is not always available in structured electronic form. Research on photographs of printed medical records has shown that optical character recognition (OCR), combined with document interpretation, can extract machine-readable information and support the construction of structured personal health records [4]. Earlier clinical OCR research also demonstrated the feasibility of capturing structured information from paper-based clinical forms while highlighting the importance of accuracy and suitability within real-world workflows [8].

HealthOrbit incorporates a medical-document processing component that enables users to submit supported medical documents for OCR-based text extraction. The document-processing workflow converts the textual content of supported documents into machine-readable information that can be further handled and presented through the application. The approach provides a practical mechanism for making information contained in scanned or image-based medical documents more accessible within a unified health-information environment.

C. Artificial Intelligence and Conversational Health Systems

Artificial intelligence has been investigated across healthcare applications, including medical imaging, decision support, patient-facing systems, and human–AI collaboration [9]. Conversational agents have also been systematically studied, with applications ranging from health information and education to monitoring and patient support [5], [6]. However, evidence of effectiveness varies across applications, and recent literature emphasises the importance of systematic evaluation rather than assuming that the presence of an AI model itself establishes clinical value [6], [7].

HealthOrbit incorporates contextual AI assistance as an interaction capability within its broader health-information environment. The Health Context concept enables relevant and authorised health information to be considered when providing context-aware assistance to the user. This approach is intended to help users interact with information already available within the HealthOrbit environment rather than presenting the system as an autonomous diagnostic or treatment system. The AI-assisted functionality therefore focuses on health-information interaction and contextual support, while clinical decision-making remains outside the scope of the system.

D. Privacy, Security, and Interoperability

Health information systems require careful consideration of confidentiality, access control, consent, authentication, security mechanisms, and interoperability. Systematic reviews of electronic health records identify access control, consent, cryptography, auditability, interoperability, and scalability as important technical and organisational concerns [10], [11]. Research on cybersecurity in eHealth further highlights the vulnerabilities associated with healthcare systems and the need for appropriate evidence-based protection mechanisms [12].

HealthOrbit incorporates authenticated access as an important security boundary for the application and provides controlled interaction with user health information. Medical documents require particular attention because they may contain personally identifiable and sensitive health information. The system therefore considers secure access and controlled handling of health-related information as essential design requirements.

For interoperability, HL7 FHIR provides a standardised approach for exchanging healthcare information electronically and represents an important direction for extending HealthOrbit to external health-data sources [13]. The present system is designed with extensibility in mind, allowing future integration with standardised healthcare information-exchange mechanisms.

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E. Research Gap

The literature demonstrates established research streams in personal health records, medical-document processing, conversational healthcare agents, privacy, and interoperability. The research opportunity addressed by HealthOrbit is therefore framed primarily as an integration problem rather than as a claim that the individual technologies are novel.

HealthOrbit brings together a user-centred health-information environment, authenticated access, medical-document processing, OCR-based information extraction, and a framework for contextual AI-assisted interaction within a unified platform. The system provides an integrated approach for organising and interacting with personal health information while establishing a structured Health Context through which relevant information can be considered during user interactions.

The contribution of HealthOrbit lies in the design and integration of these capabilities within a common digital health-information environment. The platform demonstrates how document-based health information can be incorporated into a broader information-management workflow and how the resulting health context can provide a foundation for context-aware assistance.

III. PROBLEM DEFINITION

Personal health information is often fragmented across documents, applications, records, and other sources, making it difficult for users to maintain a unified understanding of information that they already possess. Medical reports may contain valuable information in document form, while other health information may be distributed across different records or interfaces. Users can therefore face unnecessary effort when locating, reviewing, organising, and contextualising their own health information.

The problem addressed by HealthOrbit is the development of a unified digital environment for personal health information in which users can securely access, organise, and interact with available health information through a common application interface. The system also provides medical-document processing and OCR-based information extraction, while its Health Context framework establishes a foundation for context-aware AI-assisted interaction with authorised health information.

IV. AIM AND OBJECTIVES

HealthOrbit aims to provide an integrated digital health information platform that simplifies access to personal health information and supports context-aware interaction with authorised health-related information.

The objectives of the HealthOrbit system are:

1) To provide a unified interface for personal health information management.
2) To provide authenticated access to the HealthOrbit application.
3) To provide medical-document management and OCR-based document processing.
4) To extract machine-readable information from supported medical documents.
5) To organise available health information within a common application environment.
6) To provide a Health Context framework for context-aware AI-assisted interaction using authorised user information.
7) To provide an extensible architecture for health-data integration, visualisation, and intelligent assistance.
8) To support secure and user-oriented interaction with personal health information.

V. DEVELOPED HEALTHORBIT SYSTEM

HealthOrbit is an integrated digital health information platform designed to provide a unified environment for managing and interacting with personal health information. The system brings together authenticated application access, health-information management, medical-document processing, OCR-based information extraction, and contextual assistance within a common platform. The application is organised around the user's interaction with HealthOrbit. The user first accesses the system through the authentication interface, after which authenticated access is provided to the available health-information services. Users can interact with supported health-information modules and submit medical documents through the document-processing interface.

When a supported medical document is uploaded, the document is received by the application and processed through the document-processing pipeline. OCR is applied to extract machine-readable textual information from the document. The extracted information is then handled by the application and made available for further processing, organisation, or presentation according to the corresponding HealthOrbit functionality.

The broader HealthOrbit platform provides an integrated structure for health information such as medical reports, health history, medicines, symptoms, measurements, lifestyle information, and other user-provided information. The Health Context framework connects relevant authorised information with contextual assistance, allowing the system to provide a more information-aware interaction environment.

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The system is designed as a modular platform so that individual components can operate together while also allowing additional health-information sources and services to be integrated. This architecture supports the future expansion of HealthOrbit while maintaining a common interface for users to access and interact with their personal health information.

VI. SYSTEM ARCHITECTURE

Fig. 1 presents the system architecture of HealthOrbit and illustrates the interaction between the user interface, application server, medical-document processing, data management, analytics and AI assistance, health-management services, and output components. The architecture follows a layered design in which user requests are received through the client layer and processed by the application server. The application server manages request routing, authentication, user sessions, business logic, and communication with the underlying services. Medical documents are handled through a dedicated processing pipeline that includes document upload, file validation, preprocessing, OCR-based text extraction, information extraction, and data structuring. The extracted information is subsequently transferred to the data-management layer, where health information can be organised, stored, and associated with the corresponding user's Health Context. This separation allows document processing to remain independent from the presentation layer while enabling the extracted information to be reused by other platform components. The data-management and storage layer acts as the central information-handling component of the platform. It provides mechanisms for organising health-related information and maintaining the relationship between stored information and the corresponding user. The architecture also provides interfaces through which external data sources and services can exchange information with the application. These include document-processing services, health-data platforms, AI services, and notification services where applicable. The analytics and AI assistance layer operates on authorised health information available within the platform. It supports health-context construction, information analysis, trend identification, and context-aware interaction. The Health Context provides a logical connection between user information and the assistance layer so that relevant information can be considered when responding to user interactions. The AI component is intended to support health-information interaction and does not replace professional medical diagnosis or treatment. The health-management layer provides an organised view of information such as medicines, symptoms, measurements, nutrition and lifestyle information, emergency information, insurance information, and family-related access. The output layer converts processed information into user-facing representations such as dashboards, reports, trend visualisations, AI-assisted responses, and notifications. Security and privacy mechanisms apply across the architecture, with authentication and controlled access providing the primary boundary for user-specific information. The modular organisation of the architecture allows individual components to communicate through defined interfaces while keeping their responsibilities separated. This structure improves maintainability and provides a basis for integrating additional health-data sources, services, and intelligent capabilities without changing the fundamental interaction model of the platform.

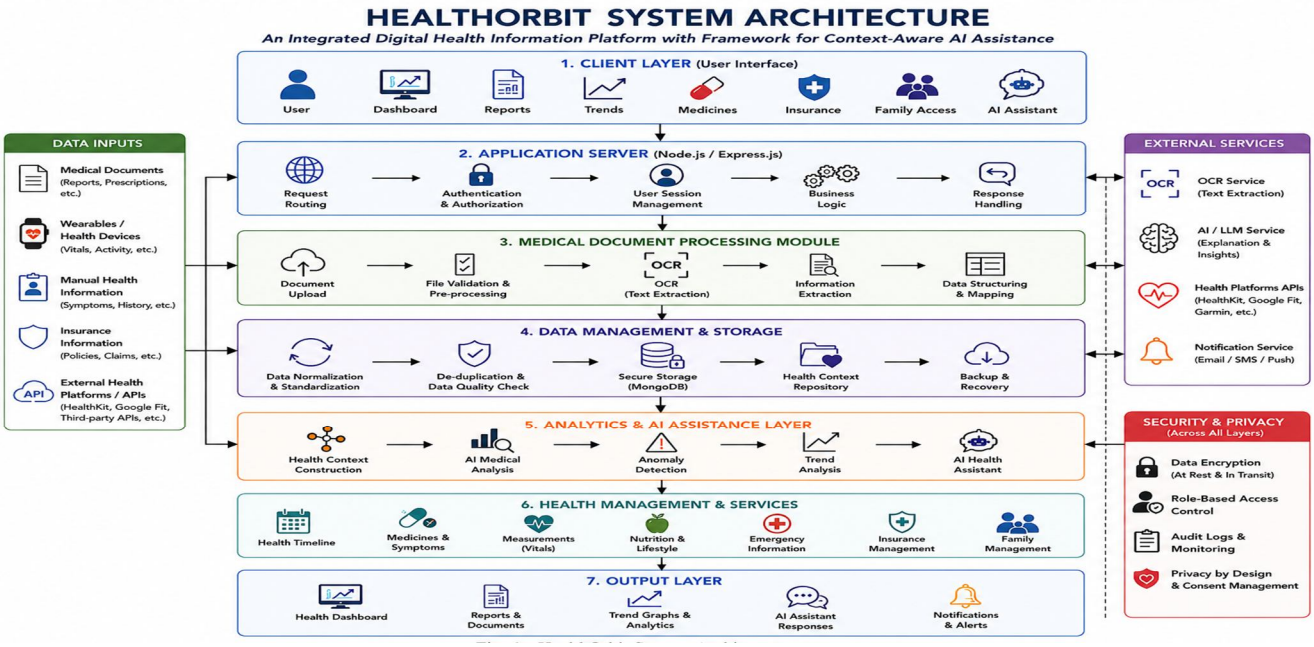


Fig. 1. HealthOrbit System Architecture

The architecture is intentionally modular. Authentication controls access to the application; the application layer coordinates user requests; persistent storage manages available information and documents; the document-processing component handles OCR-related operations; and the Health Context/AI layer provides a foundation for future contextual assistance. The diagram does not assert the implementation of future services such as wearable integrations or retrieval-augmented generation.

VII. SYSTEM MODULES

HealthOrbit is organised into interconnected modules that collectively support personal health-information management, medical-document processing, contextual assistance, and information presentation. Table I summarises the principal modules and their corresponding functions within the HealthOrbit platform.

Table I. Healthorbit System Modules And Functions

Module	Core Function
Login / Authentication	Provides authenticated entry into the HealthOrbit application.
User Profile	Provides the conceptual location for personal health information and preferences.
Medical Reports	Provides access to the medical-document functionality.
OCR / Document Processing	Processes supported medical documents and extracts machine-readable information.
Health Context	Organises relevant authorised health information for contextual interaction.
Health Timeline	Organises longitudinal health information for review.
Health Trends	Supports visualisation and interpretation of measurements over time.
AI Medical Analysis	AI-assisted analysis of health information.
AI Health Assistant	Conversational health-information assistance using Health Context.
Medicines & Symptoms	Management of medicine and symptom information.
Wearable / Health Data	Integration of device-derived health information.
Emergency Information	Access to emergency-related health information.
Insurance Management	Organisation of insurance-related health information.
Nutrition & Lifestyle	Management of nutrition and lifestyle information.
Dashboard and Reports	Presents organised health information through dashboards, reports, and visualisations.
Family Management	Supports authorised sharing or access to relevant health information.

The modular organisation allows individual functions to operate within a common application environment while maintaining separation between user interaction, document processing, data management, intelligent processing, and information presentation. This modular structure also supports integrating additional health-information sources and services without changing the platform's overall organisation.

VIII. WORKING OF THE SYSTEM

The working of HealthOrbit begins when an authenticated user accesses the medical-document module through the application interface. The user selects a supported medical document and submits it for processing. The application backend receives the uploaded file and performs the required file handling and validation before passing it to the document-processing module.

The document is then processed using the OCR component to extract machine-readable text from its contents. The extracted information is passed to the information-extraction stage, where relevant textual content is organised into a form the application can handle. The processed information can subsequently be stored or presented to the user through the HealthOrbit interface.

Fig. 2 illustrates the implemented data flow from document submission to user review. The workflow separates document upload, backend processing, OCR, information extraction, and information presentation, allowing each stage to perform a defined function within the system.

The overall workflow can be represented as:

User → Authentication → Document Upload → Backend Processing → OCR → Information Extraction → User Review

The extracted information remains machine-processed and may depend on document quality, layout, image resolution, and text clarity. Therefore, the user can review the extracted information before using it for further health-information management or interpretation.

The workflow also maintains a clear separation between the original uploaded document and the information produced during processing. The original document can be retained with its associated metadata, while the extracted text is handled separately for review and further application-level processing. This separation helps preserve the source document and provides traceability between the uploaded file and its extracted information. The user remains responsible for reviewing the extracted content before using it for subsequent health-information management.

From a system-design perspective, the workflow is modular, allowing document upload, backend processing, OCR, information extraction, and presentation to operate as distinct processing stages. This structure makes the document-processing pipeline easier to maintain and provides a basis for extending HealthOrbit with additional document types, structured health records, and other information-management capabilities. Any such extensions would require implementation and evaluation before being considered part of the verified system functionality.

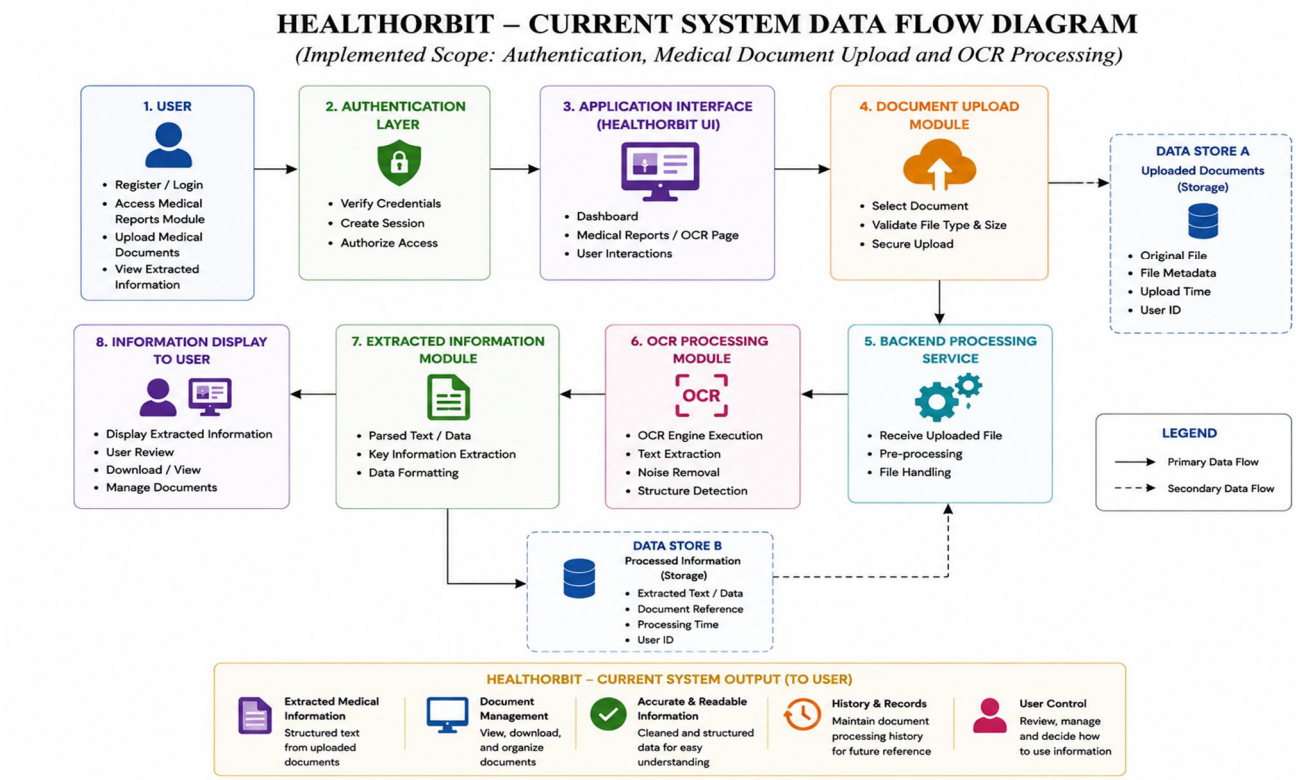


Fig. 2. Current Data Flow in HealthOrbit (Implemented Scope)

IX. MEDICAL DOCUMENT AND OCR PROCESSING

Medical document processing is a significant component of HealthOrbit. The workflow accepts a supported document from the user, performs file handling and OCR-based text extraction, and makes the resulting information available through the application. OCR is particularly useful when medically relevant information is contained in scanned or image-based documents rather than already structured electronic data.

The research literature supports the feasibility of converting medical-document images into machine-readable information. For example, previous work on printed medical records has combined OCR with document interpretation and information extraction to construct structured personal health records [4]. Earlier clinical OCR research also demonstrated the feasibility of extracting structured information from paper-based clinical forms while highlighting the importance of accuracy and suitability within real-world workflows [8].

In HealthOrbit, OCR acts as the processing stage that bridges the original document and machine-readable information. The quality of the resulting text depends on characteristics of the input document, including image resolution, document layout, scanning quality, typography, tables, and medical terminology. Consequently, OCR output is treated as machine-extracted information and should be reviewed by the user before being used for health-related interpretation.

X. HEALTH CONTEXT AND AI ASSISTANCE

Health Context is the conceptual representation of health information that may be made available to HealthOrbit's contextual assistance layer. It can include information such as a user's health profile, medical reports, medicines, symptoms, health measurements, historical information, and previous conversations, where such information is stored and authorised for use.

The current project evidence supplied for this document confirms the medical-document/OCR workflow but does not provide source-code evidence for a fully operational context-retrieval or retrieval-augmented generation pipeline. Therefore, contextual AI assistance is described here as an architectural capability and future development direction rather than as a measured, clinically validated feature.

The intended interaction model is:

User Question → Authentication → Relevant Health Context → Context Construction → AI Processing → Response → User

Future implementation should ensure that only authorised information is supplied to the AI service and that responses are handled with appropriate safety controls.

Health history and AI conversation history should remain conceptually distinct. Health history represents information about the user's health over time, whereas conversation history represents the interaction record between the user and the AI system.

XI. TECHNOLOGIES USED

HealthOrbit is implemented as a web-based application using a modular client-server architecture. The frontend provides the user interface and document interaction, while the backend manages authentication, request processing, document handling, and communication with persistent data services. The following technologies constitute the implementation stack used in the current project.

TABLE II HEALTHORBIT IMPLEMENTATION TECHNOLOGIES

Layers	Technology / Tool	Purpose
Frontend	HTML5, CSS3, JavaScript, EJS	User interface, page rendering and interaction
Backend	Node.js, Express.js	Server-side processing, routing and API handling
Database	MongoDB, Mongoose	Persistent storage and database interaction
Authentication	Express Session, Google OAuth	User authentication and session management
Document Processing	Tesseract.js, PDF processing	OCR and extraction of text from supported documents
File Handling	Multer / Cloudinary	Medical-document upload and storage
AI Service	Google Gemini / GenAI / Mesh API	Planned integration for future contextual AI assistance

The technologies listed above correspond to the current implementation of the HealthOrbit prototype. Components identified as future extensions are explicitly distinguished from the implemented functionality.

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XII. HEALTHORBIT DATA FLOW

Fig. 3 presents the proposed logical flow of health information across the broader HealthOrbit platform. The model illustrates how information from sources such as medical documents, manually entered health information, and future external health-data sources may be collected, validated, processed, stored, and made available for Health Context construction and contextual assistance. The figure represents the extensible data-flow model of the platform and should not be interpreted as evidence that all depicted integrations or AI components are currently implemented.

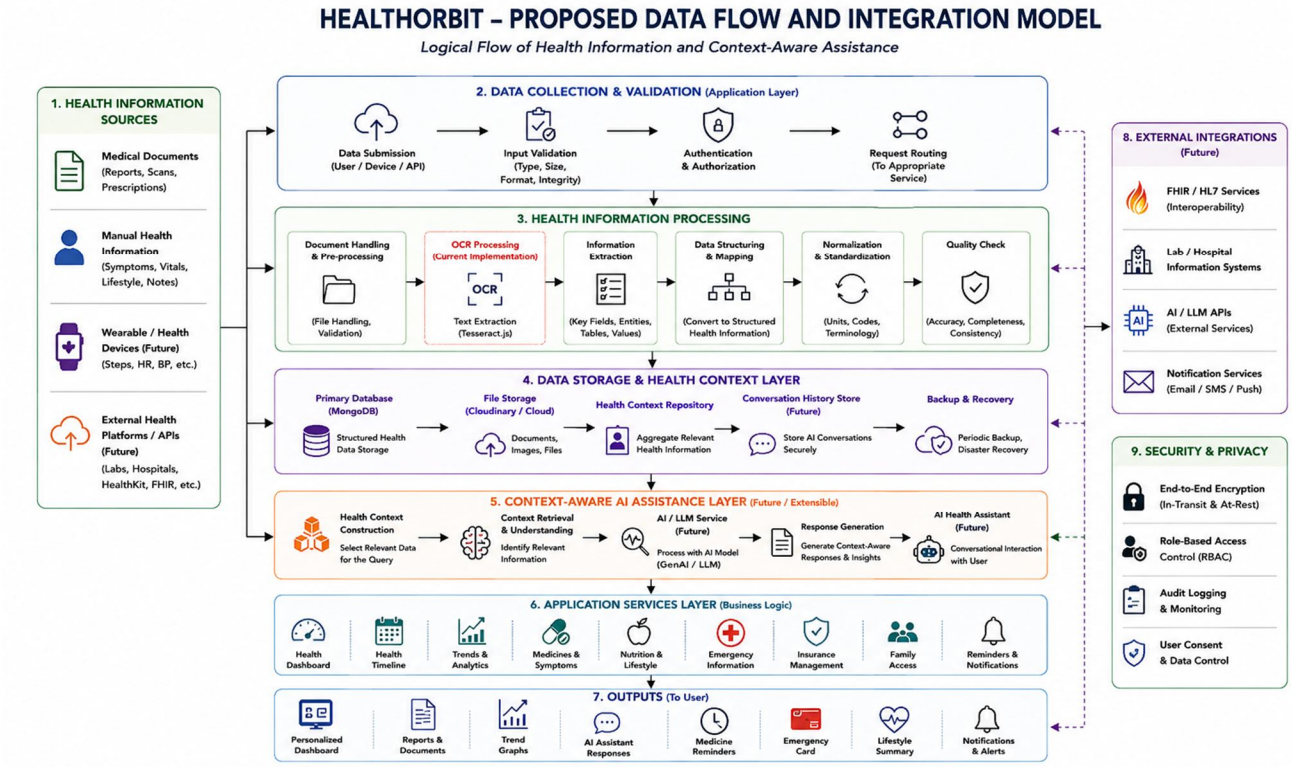


Fig. 3. HealthOrbit Data Flow and Integration Architecture (Proposed Model)

The implemented medical-document pathway forms the concrete basis of this model. In the current system, a supported medical document is submitted by an authenticated user, processed through the backend and OCR pipeline, and converted into machine-readable information for subsequent presentation or application-level handling. Broader health-data integration, automated Health Context retrieval, and contextual AI processing represent planned extensions that require further implementation and evaluation.

XIII. SECURITY AND PRIVACY

HealthOrbit handles health-related information and therefore requires careful consideration of confidentiality, integrity, availability, authorisation, and user control. Research on electronic health records repeatedly identifies access control, authentication, consent, auditability, cryptography, and interoperability as important security and privacy concerns [10], [11]. Recent research also emphasises the privacy, security, and reliability risks that can arise when AI is integrated into healthcare information systems [12], [14].

The implemented authentication mechanism provides an access-control boundary for the HealthOrbit application. Medical documents also require secure handling because they may contain personally identifiable and sensitive health information. In an expanded deployment, access to stored documents and health information should be restricted to authorised users, data transmission should be appropriately protected, and third-party AI services should be evaluated with respect to their data-handling and privacy practices.

HealthOrbit is not represented in this paper as HIPAA-compliant, GDPR-compliant, clinically validated, or otherwise certified under a healthcare regulatory framework. Such claims require formal assessment and supporting evidence. Privacy-preserving design, explicit consent, auditability, secure data handling, and appropriate data-governance mechanisms are therefore identified as important areas for continued development.

XIV. RESULTS AND DISCUSSION

The current evaluation is primarily implementation-oriented. The available project evidence confirms a functioning authentication/login interface and an implemented medical-document/OCR workflow. These components demonstrate the application's ability to provide authenticated access and process supported medical documents through an OCR-based pipeline.

The document-processing workflow provides a practical mechanism for converting information contained in supported medical documents into machine-readable information that can be handled by the application. This is consistent with prior research showing that OCR-based approaches can support the transformation of printed or scanned medical information into machine-readable personal health records [4], [8].

No numerical accuracy, precision, recall, F1 score, response latency, user-satisfaction score, clinical outcome, or large-scale user-study result is reported because corresponding experimental measurements were not available in the project materials used for this study. Such evaluation should be performed in future work using representative medical-document datasets, controlled functional test cases, usability studies, document-extraction metrics, and appropriate AI-response and safety assessments if the contextual AI assistant is implemented.

The implemented components provide a foundation for the broader HealthOrbit architecture. The principal remaining challenge is to extend the platform with additional health-information sources and contextual assistance while maintaining privacy, reliability, data integrity, and clear boundaries between health-information support and clinical decision-making.

XV. LIMITATIONS

1) The current verified implementation scope is narrower than the complete HealthOrbit vision, and several health-management, integration, and AI capabilities remain planned or extensible components.
2) OCR performance can vary depending on document quality, layout, image resolution, scanning quality, handwriting, tables, typography, and medical terminology.
3) The available project materials do not include a formal quantitative evaluation dataset, benchmark results, or user-study results.
4) The contextual AI architecture requires further implementation and evaluation before claims about personalised or context-aware AI assistance can be made.
5) Clinical validation has not been established; therefore, HealthOrbit should not be treated as a diagnostic or treatment system.
6) Broader interoperability with external health-data sources and healthcare information standards remains future work.

XVI. FUTURE SCOPE

Future development can extend HealthOrbit from the currently implemented authentication and medical-document/OCR workflow into a broader personal health-information ecosystem. The major areas of future development include:

1) Wearable and health-device integration, including standardised health-data interfaces.
2) Health timeline and longitudinal trend visualisation.
3) Structured management of medicine, symptom, nutrition, and lifestyle information.
4) A context-aware AI Health Assistant with controlled retrieval of authorised health information.
5) Retrieval-augmented generation where justified by the application's data, architecture, and evaluation requirements.
6) AI-assisted anomaly and trend analysis, subject to appropriate validation and safety controls.
7) Emergency and insurance information modules.
8) Multilingual and voice-based interaction.
9) Stronger consent, audit, privacy, and data-governance mechanisms.
10) Formal usability, reliability, document-extraction, AI-response, and security evaluation.
11) Interoperability with standards such as HL7 FHIR for future health-data exchange [13].
12) Clinical collaboration and validation before any clinical-use claims are considered.

XVII. CONCLUSION

HealthOrbit presents an integrated approach to personal health-information management that combines authenticated application access with medical-document processing and an extensible architecture for contextual health-information assistance. The implemented authentication and OCR-based document-processing workflow provides a concrete foundation for the platform, while the Health Context concept establishes a structured direction for future context-aware AI-assisted interaction.

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The project's contribution should be understood as the design and development of an integrated health-information environment rather than a claim of clinical effectiveness or universal novelty. Personal health records, OCR-based medical-document processing, conversational agents, privacy and security engineering, and healthcare interoperability are established areas of research. HealthOrbit brings these areas together within a common platform architecture and provides a practical foundation for further development and integration.

Future work should focus on expanding the implemented health-information modules, implementing and evaluating contextual AI assistance, strengthening privacy and security controls, and conducting systematic technical, usability, and document-processing evaluation. Clinical collaboration and validation would be required before making any claims regarding diagnostic, treatment, or other clinical utility.

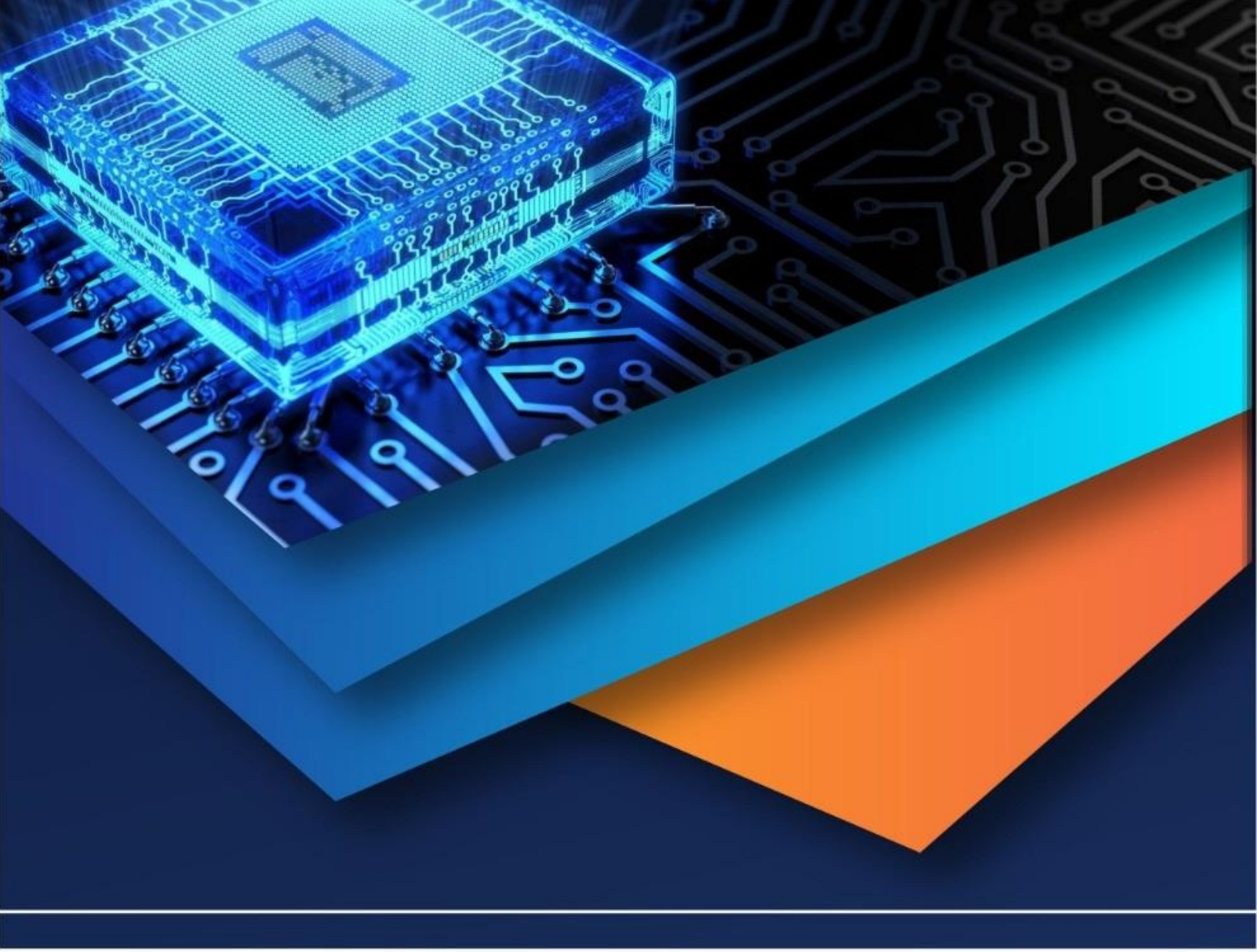
XVIII. ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to Prof. K. R. Sarode for his valuable guidance and support throughout the development of the HealthOrbit project. The authors also acknowledge the Department of Computer Engineering, Smt. Radhikatai Pandav College of Engineering, Nagpur, for providing the academic environment and resources necessary for this work.

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